

Octavio Villalobos

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EDUCATION

University of California, Santa Cruz <i>Master of Science in Computer Science — GPA: 3.74/4.00</i>	June 2026 Santa Cruz, CA
University of California, Santa Cruz <i>Bachelor of Science in Computer Science, With Honors — GPA: 3.60/4.00</i>	June 2024 Santa Cruz, CA

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, C#, SQL, HTML/CSS, Angular
Front-end Development: React.js, React Native (Expo), Tailwind CSS, HTML5
Backend & Databases: FastAPI, Express.js, Node.js, RESTful APIs, PostgreSQL, Supabase
AI/ML & Generative AI: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Hugging Face, OpenAI API, Claude API, LoRA, QLoRA, Ollama, ChromaDB, MLflow, Weights & Biases (W&B), Pandas, Matplotlib, Seaborn
Cloud & Infrastructure: Google Cloud Platform (GCP), AWS, Docker, Linux
DevOps & Testing: Git, GitHub Actions, CI/CD Pipelines, Pytest

EXPERIENCE

Research Project Lead @ Sloth Lab – UCSC <i>Santa Cruz, CA</i>	Oct 2025 – Present
– Led a 3-person team extending the Lyra opinion dynamics framework into a scalable multi-agent simulation system for personality-driven NPC interactions and emergent social behavior – Designed agent architectures with 23-dimensional state tracking and prospect-theory-based persuasion modeling, implementing per-agent Actor-Critic (A2C) RL policies across 5 personality archetypes – Developed clustering and behavioral-analysis pipelines using MLflow experiment tracking, validating emergent interaction patterns across 1,000+ large-scale simulation runs – Publishing a research paper submitted to AIIDE 2026 exploring emergent social behavior, conversational dynamics, reinforcement learning, and multi-agent interaction systems	
AI Engineer Intern @ Silicon Valley Certification Hub <i>Palo Alto, CA</i>	June 2025 – Present
– Developed and deployed an LLM-powered coaching platform (Flask, AWS) built around a RAG-based retrieval pipeline that generates executive-style scenario assessments and coaching feedback – Engineered rubric-based scoring pipelines using structured prompt engineering and configurable grading curves to evaluate answer relevance, ethical risk coverage, and actionability – Fine-tuned Gemma-3-1B and LLaMA-3.2-1B via LoRA (PEFT), improving factual alignment up to 25% and response consistency up to 9% on held-out eval data – Implemented role-conditioned scenario generation and optional ElevenLabs text-to-speech integration, supporting realistic AI interview simulations across 12 industries	
Teaching Assistant @ University of California, Santa Cruz <i>Santa Cruz, CA</i>	Sep 2024 – June 2026
– Delivered lectures on databases, AI, LLMs, and technical writing to 150+ students across 4 UCSC courses – Led weekly discussion sections for 30 students, developing instructional materials and hands-on lesson plans – Designed coursework, assignments, and exams across 6 courses to improve alignment between lectures	

PROJECTS

FormulaForecast <i>Python, Pandas, NumPy, PyTorch, scikit-learn</i>	Nov 2025 – Jan 2026
– Developed a deep learning framework in PyTorch for predicting Formula 1 top-10 race finishing positions, leveraging historical race telemetry and driver-performance data from the FastF1 API – Engineered 9 domain-specific features, implementing StandardScaler normalization, 3-sigma outlier clipping, and filtering pipelines to remove DNFs and high-variance race incidents – Designed and trained a 3-layer MLP neural network (128→64→32) using ReLU activations, Dropout, Batch Normalization, and He/Kaiming initialization	

- Achieved an MAE of 1.585 and 88.7% within-3 prediction accuracy on unseen 2025 race data while reducing prediction error by 41% compared to baseline full-grid neural network models

Insight Deck | *Python, React, Flask, LangChain, Vector DB, ChromaDB, RAG*

Apr 2025 – Dec 2025

- Co-developed a time-aware hybrid retrieval pipeline combining **BM25**, **ChromaDB** vector search, and metadata-aware reranking, improving form-type retrieval accuracy across multi-year financial corpora
- Orchestrated a multi-agent **RAG** workflow using **LangChain**, coordinating Search, Parser, Critique, and Slide Generation agents to automate investor presentation creation from scraped financial and company data
- Engineered **dynamic query-generation** and **prompt-planning** pipelines with overlapping Markdown chunking, semantic retrieval, and contextual refinement workflows, improving content population from 74.2% to 98% while reducing slide-planning latency by 49%
- Reduced slide-generation failures and low-content outputs by engineering an iterative **LLM critique** and repair pipeline from 70% to 90% and improving presentation completeness by 10–15% per iteration

LLMs are ACDs | *Python, CybORG, CAGE 4, OpenAI API, HuggingFace, Ollama*

Apr 2025 – Dec 2025

- Co-authored an **IEEE CAI 2025** paper developing the first framework for **LLM** and **RL** agents in **CybORG CAGE 4** multi-agent cybersecurity environment, enabling interpretable autonomous cyber defense simulations
- Collaborated within a 6-person research team to create an 8-bit communication protocol enabling coordination across agents (**LLM + RL**), supporting real time threat sharing and distributed defense across 4 network zones
- Benchmarked **OpenAI**, **Ollama**, and **Deepseek** against **RL** baselines across 5 attacking agents, applying behavioral analysis and reward-based evaluation to measure interpretability, action quality, and defensive performance
- Built **Python** adapters integrating **OpenAI**, **DeepSeek**, and **HuggingFace** APIs into **CybORG**, optimizing LLM action-selection workflows to support scalable 500-step adversarial simulations; RL agents achieved $\sim 104\times$ faster action selection (45s vs. 4,705s per episode)

PUBLICATIONS

LLMs are Autonomous Cyber Defense Agents | *IEEE Conference on Artificial Intelligence (CAI 2025)*

- IEEE Conference on Artificial Intelligence (CAI 2025) — arXiv:2505.04843